

Reading and Sketching Slope Fields

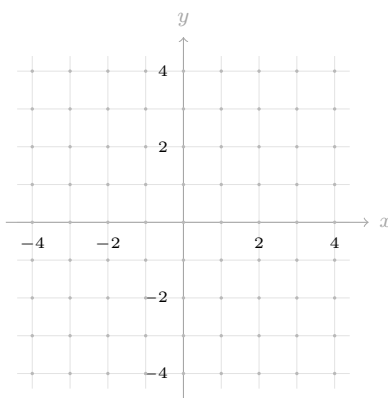
Slopes at lattice points, reading a printed field, and sketching the solution curve through a point

Directions. A slope field draws, at each lattice point, a short segment whose slope is the value the differential equation gives there. Work each problem in this order:

- **Substitute** the point's coordinates into the right-hand side — that number *is* the slope of the segment at that point.
- **Look for structure** before drawing. If the right-hand side has no y in it, every segment in a vertical column is parallel; with no x , every segment in a horizontal row is parallel. Setting the right-hand side equal to zero gives the curve along which every segment is horizontal.
- **Follow the flow** to sketch a solution curve: start at the given point and move so that the curve stays tangent to the nearby segments, never crossing one at an angle.

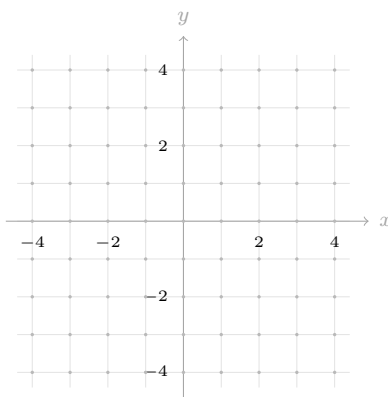
1. $\frac{dy}{dx} = x - y$. Complete the slope table, then draw a short segment of the right slope at each of those four points on the grid.

Point	(1, 0)	(2, -1)	(-1, 2)	(0, 2)
Slope				



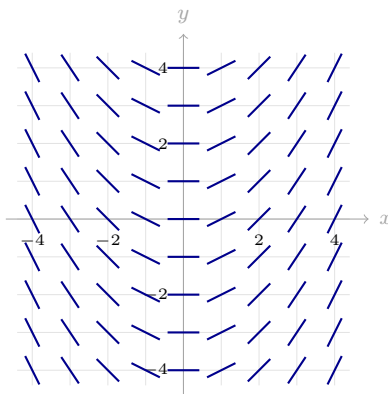
Answer: _____

2. $\frac{dy}{dx} = 2 - y$. (a) Find the slope at every point whose height is $y = 0$, then at every point with $y = 3$, then at every point with $y = 4$. (b) Solve $2 - y = 0$ to find the horizontal line made entirely of horizontal segments — the constant solution — and draw it on the grid. (c) Draw the field along those three heights.



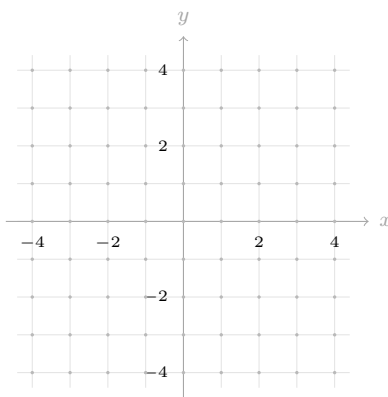
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3. The field below was generated by exactly one of $\frac{dy}{dx} = \frac{x}{2}$, $\frac{dy}{dx} = \frac{y}{2}$, $\frac{dy}{dx} = x - y$. (a) Read the slope off the field in the column $x = 4$ and again in the column $x = -2$. (b) Solve the winning right-hand side equal to zero to say exactly where the horizontal segments are. (c) Write a sentence naming the equation and saying what rules each of the other two out.



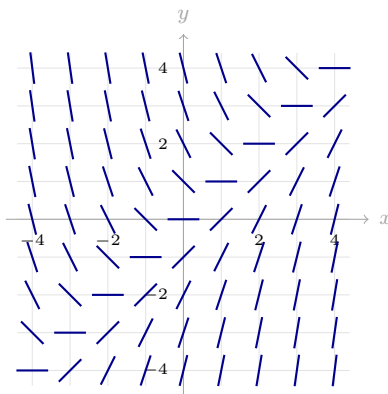
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4. $\frac{dy}{dx} = 2x$ with the curve passing through $(1, 2)$. (a) Draw the field at the lattice points shown — note that the slope never depends on the height. (b) Sketch the solution curve through $(1, 2)$. (c) Antidifferentiate $2x$, fix the constant from the given point, and check the equation agrees with your sketch by evaluating it at $x = 0$.



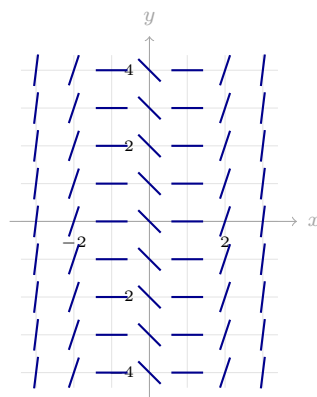
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5. Below is the finished field for the equation you started in problem 1, $\frac{dy}{dx} = x - y$. (a) Read the slope off the field at $(2, 2)$ and at $(3, 0)$, then confirm each from the equation. (b) Solve $x - y = 0$ for y : that line carries every horizontal segment. (c) Sketch the solution curve through $(0, 3)$ and describe in a sentence what it does as x grows large.



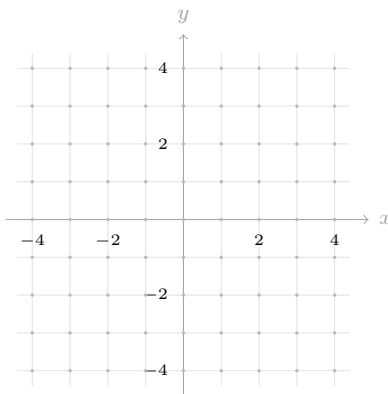
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- 6.** The field below belongs to $\frac{dy}{dx} = x^2 - 1$. **(a)** Sketch the solution curve through $(0, 1)$, using the two columns of horizontal segments to place its turning points. **(b)** Antidifferentiate $x^2 - 1$ and fix the constant from that point. **(c)** Evaluate your particular solution at $x = 2$, rounded to two decimal places, and check the value against your sketch.



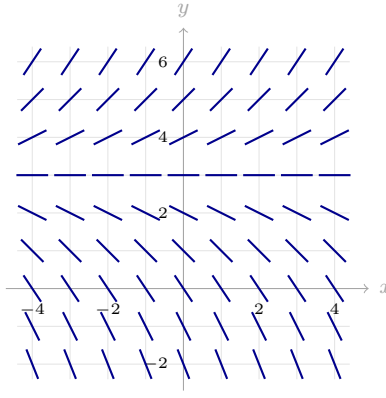
Answer: _____

- 7.** $\frac{dy}{dx} = x + y$. **(a)** Find the slope at $(1, 1)$, at $(2, -2)$ and at $(-3, 1)$. **(b)** Solve $x + y = 0$ for y to get the line every horizontal segment sits on, and draw that line on the grid. **(c)** Draw the field at the nine lattice points nearest the origin, then say which single segment on the whole grid is steepest and why.



Answer: _____

8. The field below belongs to $\frac{dy}{dx} = \frac{y-3}{2}$. **(a)** Solve $\frac{y-3}{2} = 0$ to find the equilibrium (constant) solution and draw it. **(b)** Find the slope at height $y = 5$ and at height $y = 1$, and say what the signs tell you. **(c)** Sketch the solution curves through $(0, 5)$ and through $(0, 1)$, then explain — using the field alone, not a formula — why neither curve can ever reach the equilibrium line.



Answer: _____